Mass Exit Across Competing Rulers

A Behavioral Foundation for Theories of State Development*

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Abstract. Many influential theories of state development assume that political fragmentation enables subjects to exit across competing rulers, thereby disciplining extraction, yet direct evidence that fragmentation shapes how subjects resist remains scarce. I show this in Tokugawa Japan (1615–1868), where nearly 300 feudal domains generated sharp variation in villages' proximity to rival rulers. In a difference-in-differences design, the shogunate's conversion of domains to direct rule raised the probability of nearby mass exit by 4 percentage points with no rise in direct confrontation, and a grid-year panel confirms the pattern. Among revolting villages, greater access to outside rulers tilted resistance toward exit over confrontation. Domains more exposed to credible exit threats levied lower taxes, especially where neighboring domains could absorb migrants. The findings microfound a premise that load-bears across the Great Divergence literature, fiscal-federalism arguments, and the cartel theory of the territorial state system.

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1. Introduction

Many influential theories of state development rest on a common premise: political fragmentation enhances the mobility of subjects across competing rulers, and that mobility disciplines extraction and shapes institutions. This premise carries the central causal weight of leading explanations for the Great Divergence (Cox 2017; Fernández-Villaverde et al. 2023; Jones 2003; Mokyr 2017); for accounts linking Europe's polycentric structure to constrained taxation (Abramson 2017; Dincecco and Wang 2018; Sng and Moriguchi 2014; Stasavage 2016); for fiscal-federalism arguments that competing jurisdictions tame Leviathan (Qian and Weingast 1997; Rodden 2003; Tiebout 1956; Weingast 1995); and for recent theories positing that the modern territorial state system itself arose as a cartel against subject mobility (Acharya and Lee 2018; 2023). In each case, the mechanism is the same: fragmentation offers subjects credible exit options, and the threat or use of exit forces rulers to compete or cooperate. Yet the behavioral foundation of this chain has yet to be directly demonstrated: that the intensity of jurisdictional conflict actually shaped how subjects resisted, or put differently, that subjects rationally exploited fragmented polity to check oppression. I show just this.

Direct evidence supporting this premise is scarce for two main reasons. First, data on exit are limited (Bailey 2014, 76): unlike open rebellion, which leaves abundant documentary traces, exit is often subtle and covert. As a result, most existing research infers exit indirectly rather than observing it directly at the community level. Second, it is difficult to find usable variation in jurisdictional conflict. Macro-level studies that adopt this premise typically compare entire regions, such as Europe and China, whose polities differ across many dimensions, providing little within-case variation suitable for testing a behavioral mechanism at the level where subjects actually make choices. This paper addresses both challenges.

Tokugawa Japan (1615–1868) provides a clean laboratory for this mechanism. Nearly 300 feudal domains formed a patchwork of autonomous units, each with its own legal code, tax system, and border controls, held together by the Tokugawa shogunate as hegemon, a hybrid regime of domain autonomy combined with supra-domain arbitration, attainder power, and direct peasant-appeal authority retained at the center (Moore 1993; Ravina

1999; White 1995). Three features make the setting exceptional. First, the post-1615 peace shuts off the war-finance channel that Rosenthal and Wong (2011) identify as overwhelming the exit-discipline mechanism in early modern Europe, isolating the mechanism itself (Moore 1993). Second, intense inter-domain competition for agricultural labor made mass migration a credible peasant strategy, and villages varied widely in their geographic exposure to rival rulers. Third, newly digitized records of peasant uprisings compiled by Aoki (1975) document both direct confrontations and mass exits at the village level across the entire period.[3]

The empirical analysis proceeds in three parts. First, access to alternative authority raised the *occurrence* of exit. A difference-in-differences design exploits an exogenous shock to local outside options: the shogunate's periodic abolition of domains and conversion of their territories into direct shogunal rule (*tenryo*), widely perceived as less heavily taxed and less coercively governed than daimyō domains. Tenryo conversion raised the probability of nearby exit by 3.9 percentage points while rates of direct confrontation remained unchanged, and event studies show no pre-trends and a post-conversion spike in exit. A grid-year panel tracking within-cell shifts in the distance ratio as nearby domain capitals are established, abolished, or relocated corroborates this occurrence result: greater geographic exposure predicts a higher frequency of exit events across the full panel of cells, without affecting the frequency of direct confrontation.

Second, among villages that did revolt, greater geographic exposure tilted the chosen form of resistance toward exit over direct confrontation. I measure each village's exposure as the ratio of its distance to home-domain capital to its distance to the nearest outside administrative center. Across all recorded revolts, this exit-*share* result holds robustly under home-neighbor dyad fixed effects. Decomposing the ratio into its two absolute distances reveals that proximity to a foreign capital is the active margin, confirming that

[3]Tokugawa mass exit operated as active resistance rather than permanent relocation: peasants presented demands to home-domain officials while seeking permission from neighboring rulers to cross the border, frequently returning once concessions were granted. Moore (1993, p. 255), drawing on Borton's catalog, recorded 106 such mass desertions across the Tokugawa period as evidence of "the solidarity of the Japanese village"; my catalog substantially extends this count and covers the full archipelago. The same active form appears in medieval Europe (Bailey 2014), precolonial Southeast Asia (Adas 1981; Scott 1985), and colonial India (Gandhi and Parel 1997, 95), suggesting the Tokugawa case is one window onto a broader phenomenon.

access to a rival center of power, not limited oversight by the home domain alone, drives villages toward exit.

Third, the exit mechanism leaves a fiscal trace at the domain level: rulers more exposed to alternative authority levied lower tax rates, an association that strengthens where neighboring domains had greater capacity to absorb labor, consistent both with a theoretical prediction that geographic exposure constrains taxation when the outside option is attractive enough to make exit credible, and with the long-standing political-economy claim that interjurisdictional competition disciplines ruler extraction. The association is robust to extensive sensitivity checks, including controls for domain size, geography, and neighboring domains' characteristics; alternative distance measures based on least-cost walking paths and border proximity; Conley standard errors accounting for spatial correlation; and formal tests for omitted variable bias.

The paper makes three core contributions. First, it provides direct, micro-level evidence for a central behavioral premise running through four traditions in political economy, each resting on the same jurisdictional-conflict-to-exit link: the Great Divergence argument that ruler competition for mobile subjects checked extraction and fueled European growth (Jones 2003; Mokyr 2017); state-formation accounts in which exit forced ruler bargains and built representation (Dincecco and Wang 2018; Stasavage 2016); fiscal-federalism claims that competition among jurisdictions tames Leviathan through taxpayer and labor mobility (Rodden 2003; Tiebout 1956; Weingast 1995); and the cartel theory of the territorial state, in which modern borders are erected specifically to suppress subject exit (Acharya and Lee 2018; 2023). The evidence substantiates the crucial first link in this shared causal chain: fragmentation shapes the mode of popular resistance, thereby producing effective constraints on rulers.

Second, the paper brings exit from the background to the center of the study of state development. In Hirschman's (1970) terms, the dominant analytical traditions are built around voice: revolution and rebellion drive regime change (Acemoglu and Robinson 2001; Skocpol et al. 1982), collective mobilization makes states (Hintze 1975; Tilly 1978), and the threat of revolt disciplines extraction (Besley and Persson 2009; Fearon 2011). When exit appears, it shapes incentives (Besley and Persson 2009; Fearon 2011) or long-run sorting (Jones 2003) without being directly observed. Historians of everyday resistance have long documented otherwise: exit often predominated over revolt (Adas 1981; Bailey 2014; Camp 2004; Diouf 2014; Kolchin 1990; Scott 1976; 1985; Sreenivasan 2004). This paper documents exit directly as a strategy of resistance alternative to open rebellion.

Third, this paper contributes the community-level evidence on *horizontal* exit (movement between competing jurisdictions) contrasting with the *vertical* exit (flight beyond state reach) that dominates much of the existing literature on exit (Carneiro 1970; Herbst 2000; Pruett 2024; Scott 2009).[4] Within studies of horizontal exit, the Tiebout tradition interprets mobility primarily as individual preference-sorting rather than collective resistance (Peterson 1992; Tiebout 1956). Other strands conceive of exit as a spur to voice (Hirschman 1993; Pfaff and Kim 2003; Schoppa 2022), analyze government-driven emigration (Lueders 2025; Michel, Miller, and Peters 2023), or infer horizontal mobility indirectly—from the persistence of indigenous identity near jurisdictional havens (Guardado and Franco-Vivanco 2025), from legal efforts to restrict peasant movement (Matranga and Natkhov 2025), or from tax-base reactions to cross-border competition (Kleven et al. 2020)—but do not directly capture community-level exit as resistance. The Tokugawa case uniquely pairs direct observation of mass community exit with the polycentric institutional environment to which macro-level theories of state development refer.

2. A Model of Exit and Fight under Jurisdictional Conflict

To derive testable predictions, I develop a model in which geographically heterogeneous villages choose between exit and direct confrontation in response to taxation by competing rulers. Villagers weigh compliance against two forms of resistance: walking away to a neighboring jurisdiction (*exit*) or staying to fight collectively (*fight*). Exit is cheaper for villages near a rival capital; fight requires same-domain coordination, so every villager who leaves is one fewer body for collective violence. Rulers anticipate these choices and

[4] Moore (1993, p. 230) characterized the late Tokugawa order as marked by "vertical and horizontal fissures… perhaps equally important," foregrounding the horizontal axis without operationalizing it; this paper supplies the community-level measurement.

restrain extraction only when the neighboring jurisdiction is attractive enough that their villagers would actually walk. Three predictions follow: (i) greater exposure tilts the composition of resistance toward exit in equilibrium, even after the ruler's offsetting tax cut has absorbed part of the exit pressure; (ii) exposure raises the unconditional frequency of exit events; (iii) rulers more exposed to credible outside options levy lower taxes, but only when those options are attractive enough to be taken.[5]

2.1. Model Setup

Consider a society consisting of two neighboring domains 1 and 2. In each domain, a ruler governs a continuum of villages of measure 1.[6] The domains are indexed by $j \in \{1, 2\}$ and villages in domain j are indexed by $i_j \in [0, 1]$. The game begins with the two rulers simultaneously setting tax rates $\tau_j \in [0, 1]$ for their respective domains under complete information about each other's primitives. After observing the tax rate, each village decides whether to comply or pursue one of two outside options: fight or exit. Let $a_{i_j} \in \{\text{comply}, \text{fight}, \text{exit}\}$ denote village i_j's action. The proportion of villages choosing action $a \in \{\text{comply}, \text{fight}, \text{exit}\}$ in domain j is represented by $\mu_j^a \in [0, 1]$. Each village generates yield $y_j > 0$, which varies by domain. Compliant villages retain $y_j(1 - \tau_j)$, their net-of-tax yield. I assume that a compliant village i_j receives a random utility shock $\varepsilon_{i_j} \in [-\delta, \delta]$ with $\delta > 0$, which smooths the ruler's objective function.

A direct fight against the ruler is modeled as a collective action problem: its success depends on the fraction of villages participating, and specifically, the probability of success is μ_j^{fight}. If the fight succeeds, each participating village enjoys the full yield y_j

[5]The model extends three literatures. Relative to Tiebout fiscal-competition setups where mobility is costless sorting (Tiebout 1956), exit here is costly and heterogeneous across villages. Relative to single-organization exit-voice models (Gehlbach 2006; Hirschman 1970), the two-ruler structure makes the cross-domain best-response an equilibrium object. The closest precursor on the ruler side is Acemoglu and Wolitzky (2011), whose cross-domain best-response I endogenize rather than parameterize. On exit's effect on collective action, the model commits to substitution (Sellars 2019) rather than the complementarity views (Clark, Golder, and Golder 2017; Hirschman 1993; Karadja and Prawitz n.d.; Pfaff and Kim 2003), matching the Tokugawa case in which fight required within-domain coordination.

[6]I treat the village, rather than each subject, as the unit of decision making, consistent with Tokugawa Japan's governance structure, where villages were the fundamental units for taxation, negotiation with lords, and resistance.

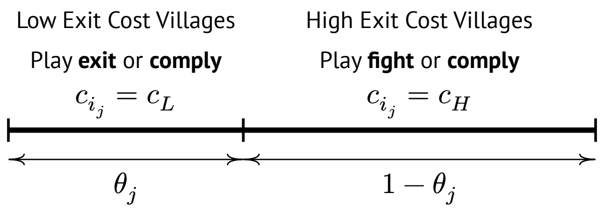


Figure 1: Exit Cost Determines Each Village's Mode of Resistance

without taxation; if it fails, they receive nothing. Choosing to fight incurs a sunk cost of $k > 0$ for each participating village.

Exit payoff is $y_{-j}(1 - \tau_{-j}) - c_{i_j}$, where $y_{-j}(1 - \tau_{-j})$ is the net-of-tax income in the other domain and $c_{i_j} \in \{c_L, c_H\}$ is the exit cost, with $c_H > c_L > 0$. The *outside-option quality* $y_{-j} > 0$ captures how attractive the neighboring domain is for potential migrants, which reflects the neighbor's economic capacity, governance conditions, and willingness to absorb newcomers.

Exit is modeled as a temporary bargaining strategy, not permanent relocation. Rather than modeling the sequential bargaining process between ruler and exiters, which would reduce tractability without additional insight, I assume rulers offer just enough concessions to induce returning villages' payoff to match the exit option, so all exiters return.[7]

The *geographic exposure* parameter $\theta_j \in [0, 1]$ denotes the share of villages in domain j that face the lower exit cost c_L; the remaining $1 - \theta_j$ face c_H. Under Assumption 1 (stated in Appendix A), high-exit-cost villages choose between fighting and compliance, while low-exit-cost villages choose between exit and compliance (see Figure 1). A greater θ_j thus means more villages with access to alternative authority.

Overall, the payoff of village i with exit cost $c_i \in \{c_L, c_H\}$ is summarized as

[7]The discussion of peasant resistance below details several historical examples of mass exits used as temporary bargaining tactics in Tokugawa Japan. Under a take-it-or-leave-it bargaining protocol in which the home ruler captures all the bargaining surplus, the village's equilibrium payoff is exactly $y_{-j}(1 - \tau_{-j}) - c_{i_j}$ as written: the ruler's concession is just enough to make the exit-threatening village indifferent, and the village's realized payoff coincides with the outside option by construction. Treating exit-threatening villages as contributing zero revenue to the home ruler is a first-order approximation (exact at the exit cutoff and mildly conservative for infra-marginal exit villages), and preserves the comparative statics of Propositions 1–3.

$$u_{i_j}\left(a_{i_j}; \tau_j, \tau_{-j}\right) = \begin{cases} y_j\left(1-\tau_j\right)+\varepsilon_{i_j} & \text{if } a_{i_j} = \text{comply} \\ y_j \mu_j^{\text{fight}} - k & \text{if } a_{i_j} = \text{fight} \\ y_{-j}\left(1-\tau_{-j}\right) - c_{i_j} & \text{if } a_{i_j} = \text{exit} \end{cases}$$

The ruler's objective is to maximize net tax revenue, taking into account losses arising from resistance. To reflect the costs associated with quelling resistance, I assume that a ruler collects taxes only from villages within her own domain that comply. Importantly, the rulers do not collect taxes from the other domain's villages exiting to her territory; as described above, such villages always return to their home domain after gaining concessions from their original ruler. The utility of domain j ruler is thus given by

$$u_j(\tau_j, \tau_{-j}) = \left[1 - \mu_j^{\text{fight}}(\tau_j) - \mu_j^{\text{exit}}(\tau_j, \tau_{-j})\right] y_j \tau_j = \mu_j^{\text{comply}}(\tau_j, \tau_{-j}) y_j \tau_j$$

for each $j \in \{1, 2\}$.

For tractability, the baseline treats exit as reversible; permanent exit (exiters absorbed into the neighbor's tax base) imposes a lasting fiscal loss on the ruler, sharpening the tax-base-versus-extraction trade-off. Appendix A.9 develops this extension and shows the main results are robust. It also treats the ruler's tenure as secure; regime-collapse risk (she loses power once revolt passes a tolerance threshold; see Section 3.2) gives her a direct incentive to cap total revolt. Appendix A.7 and Appendix A.8 develop this extension and show the main results hold.

2.2. Empirical Implications

Each proposition below is stated at interior equilibrium and maps onto a specific empirical design developed in the next section; proofs are collected in Appendix A.

2.2.1. Partial-Equilibrium Shock Response

Before tax rates adjust, a shock that makes the neighboring domain more appealing raises exit without touching fight, an asymmetric response driven entirely by where each payoff gets its arguments from.

Proposition 1 (Resistance Composition under Exogenous Shocks): Under Assumption 1 and interiority of village cutoffs, for fixed tax rates $(\tau_j, \tau_{-j}) \in (0,1)^2$: (i) $\frac{\partial \mu_j^{\text{exit}}}{\partial y_{-j}} > 0$ and $\frac{\partial \mu_j^{\text{fight}}}{\partial y_{-j}} = 0$; (ii) $\frac{\partial \mu_j^{\text{exit}}}{\partial \tau_{-j}} < 0$ and $\frac{\partial \mu_j^{\text{fight}}}{\partial \tau_{-j}} = 0$.

The exit payoff $y_{-j}(1-\tau_{-j}) - c_{i_j}$ depends on the neighbor's yield and tax rate, so any improvement there pushes more low-cost villages past the exit-comply threshold; the fight payoff $y_j \mu_j^{\text{fight}} - k$ depends only on own-domain variables, leaving the fight margin unaffected. Proposition 1 is stated at fixed tax rates to match the shock-based design in Section 3.3.1, which exogenously shifts the outside option and recovers exactly this comparative static.

2.2.2. Equilibrium Composition and Exit Rate

In equilibrium, a village's geographic exposure to a rival capital shapes both the composition of resistance and the level of exit.

Proposition 2 (Geographic Exposure and Exit): Under Assumption 1 and the regularity conditions of Appendix A:

(a) *Exit share.* Whenever $y_{-j} > \kappa_j^*$ (the threshold defined in Proposition 3 below), the equilibrium exit share among non-compliant villages,

$$\sigma_j^*(\theta_j) := \frac{\mu_j^{*,\text{exit}}}{\mu_j^{*,\text{exit}} + \mu_j^{*,\text{fight}}}$$

is strictly increasing in θ_j.

(b) *Exit rate.* The equilibrium exit rate $\mu_j^{*,\text{exit}}(\theta_j)$ is strictly increasing in θ_j.

Higher θ_j means more villages face low exit costs: the exit pool expands, the fight pool shrinks. At fixed taxes, exit rises proportionally with θ_j. The ruler anticipates this and cuts τ_j^* (Proposition 3) to keep compliance attractive, but the cut is partial: going further would sacrifice per-compliant-village revenue faster than it would recover retention. The exit rate, exit share, and total revolt therefore all rise in equilibrium.

2.2.3. Equilibrium Tax Response

Whether the ruler's equilibrium tax response to exposure is to lower or raise taxes depends on whether the neighbor's outside option is attractive enough to make exit credible.

> PROPOSITION 3 (Jurisdictional Conflict and Equilibrium Tax Rate): Under Assumption 1 and standard regularity conditions stated in Appendix A, for each $j \in \{1, 2\}$, there exists a unique threshold $\kappa_j^* > 0$ such that $\frac{\partial \tau_j^*}{\partial \theta_j} < 0$ if and only if $y_{-j} > \kappa_j^*$.

Rising θ_j shifts marginal villages from the fight margin to the exit margin. When $y_{-j} > \kappa_j^*$, exit-type villages have attractive outside options: the exit-constraint tightening dominates and the ruler lowers taxes. When $y_{-j} < \kappa_j^*$, exit is not credible and fight-pool shrinkage dominates, allowing the ruler to raise taxes: the "exit erodes voice" channel (Hirschman 1970; Sellars 2019). The empirical results are consistent with this conditional effect of jurisdictional conflict on extraction.

3. Empirical Evidence

3.1. Setting: Tokugawa Japan

During the Tokugawa period (1603–1868), Japan was divided into nearly 300 domains, each governed by a feudal lord (daimyō); Figure 2(a) maps the 264 active domain capitals in 1867, and Figure 2(b) plots the annual number of active capitals across the period as domains were created, merged, transferred, or absorbed into shogunal land. The origins of these domains lay in territories controlled by local warrior leaders during the Warring States period, or even earlier in private estates known as shoen. Formally, daimyō were subordinate to the Tokugawa shogunate, which asserted ultimate authority by establishing national policies, arbitrating inter-domain disputes, and receiving direct petitions from peasants. Yet, they continued to enjoy considerable autonomy within their own domains. In particular, each domain maintained its own government and military forces, and independently set tax rates (Ravina 1999, 1000).

Three features of this setting make it well suited for studying the relationship between jurisdictional conflict and exit as resistance. First, domains shared a broadly common administrative architecture that concentrated authority in the castle town serving as the domain capital (Hall 1955), with the village contract system (murauke) making villages directly responsible to the daimyō for tax collection (Saxonhouse 1995; Walthall 1997, 89; White 1995). This uniform, capital-centered structure justifies representing jurisdictional conflict parsimoniously by the geometry of domain capitals: a village's distances to its home capital and to the nearest rival capital capture the margin on which exit against alternative authority operated.

Second, the Tokugawa era saw no major armed conflict between domains until the shogunate's late-period collapse, marked by the Chōshū War (1864–1866) and the Boshin War (1868–1869).[8] Jurisdictional conflict among domains played out not through warfare but through competition over subjects and the "everyday forms of resistance" (Scott 1985) that this competition enabled, with external warfare playing a much smaller role than in the cases that dominate the traditional statebuilding literature (Hintze 1975; Tilly 1978; Weber 1919). This peace is a scope condition: Rosenthal and Wong (2011) theorize horizontal exit as a discipline on rulers but argue that, in early modern Europe, the mechanism was overwhelmed by war finance; the post-1615 Tokugawa peace shuts off exactly that war-finance channel, allowing the mechanism itself to be observed cleanly.

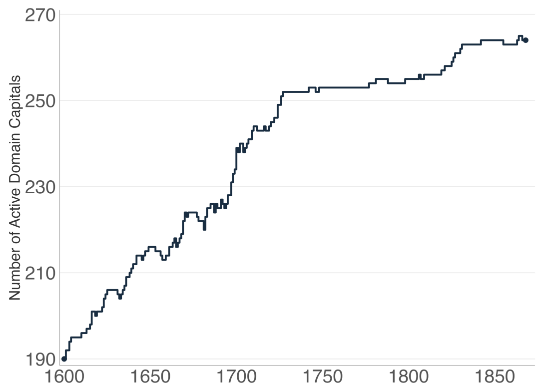
Third, the period affords an unusually rich documentary record of resistance, described below.

3.2. Peasant Resistance: Fight and Exit

Aoki's (1975) catalog identifies five forms of resistance: coercive appeals (gōso), destruc-

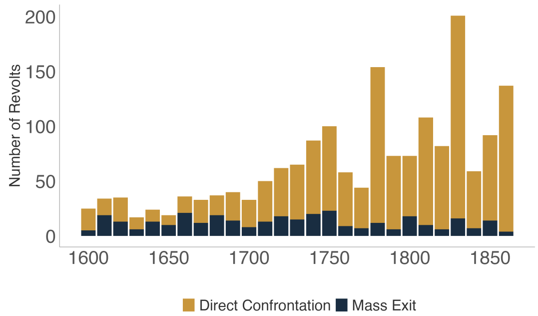
[8]The Shimabara Rebellion of 1637–38 is a mid-period exception but was an anti-Christian and anti-Matsukura uprising suppressed by a shogunal-led coalition of western daimyō, not a free-form inter-*daimyō* war. The Chōshū War was similarly a shogunate-versus-Chōshū conflict that pulled several daimyō into an ad-hoc coalition, not decentralized inter-domain warfare; the Boshin War that followed was a regime-collapse sequence. Between Shimabara and the 1860s, institutional peace held. My tax rate data are drawn from the end of the Tokugawa period and could in principle reflect Chōshū- or Boshin-era disruptions; reassuringly, historical evidence indicates that tax policies remained highly stable throughout the period, including its final years (Smith 1958).

(a) Domain Capitals in 1868 (N = 264)



(b) Number of Active Domain Capitals, 1600–1868

(c) Spatial Distribution of Peasant Revolts



(d) Temporal Trend of Peasant Revolts

Figure 2: Domain and Peasant Revolt Data

Note: (a) Active domain capitals in 1867 (black points) over Japan's terrain. (b) Annual number of active capitals, 1600–1868. (c) Revolt villages, colored by mode of resistance (exit vs. fight); gray = non-revolting villages extant through end of period. (d) Decade-by-decade revolt frequency.

tion (uchikowashi), rebellion (hōki), desertion (chōsan), and end-run appeals (osso).[9] I consolidate these into two categories: "fight" strategies (direct confrontation through coercive appeals, destruction, and rebellion) and "exit" strategies (desertions and end-run appeals that sought protection from an authority outside the home domain). Figure 2 (c) presents the spatial distribution of peasant revolts, and (d) presents the frequency of revolts over time.

The three fight subcategories differed in intensity (Aoki 1975). Coercive appeals were mass marches on domain offices or castles — crowds of thousands donning straw raincoats

[9]Subsequent studies follow his categorization: White (1995) provides a quantitative treatment of protest repertoires and their evolution; Vlastos (1986) treats end-run appeals (osso) as a distinct protest form that dominated the mid-Tokugawa period; Walthall (1997) translates primary documents from representative episodes across all five categories.

and carrying bamboo spears as ritual symbols of collective resolve (Walthall 1997, 131–132) — to present demands backed by the implicit threat of violence. Destruction involved targeted attacks on the property of officials, merchants, or wealthy villagers perceived as complicit in oppressive policies; participants organized into groups to systematically "smash" (*uchikowashi*) houses and storerooms (Walthall 1997, 89–130). Rebellion was armed uprising against domain authority, the rarest form, accounting for roughly 1% of recorded incidents (White 1995, 142). All three sometimes won concessions but often met severe repression.

The exit coding combines two routes: desertion relied on physical relocation toward a neighboring jurisdiction, whereas end-run appeals involved formal petitioning of an external authority — whether a neighboring lord or the shogunate — to intervene on the villagers' behalf. Both share the defining feature that distinguishes exit from fight: the protester seeks protection from an authority outside the home domain, so the home ruler faces the threat of losing subjects or legitimacy to an external actor.

3.2.1. Revolts Were Consequential

Both exit and fight carried the risk of attainder (*kaieki*) for the home daimyō. The Buke Shohatto (Laws of Military Houses) required domains to be "well-governed" (Steele, Paik, and Tanaka 2017, 3–5), and governance failure served as a recognized pretext for the shogunate to strip a lord of his domain, reduce his territory, or forcibly transfer him to a smaller holding. The two forms of resistance differed, however, in their *publicizing* risk. Fight typically remained a domain-internal matter: information about mass assemblies or property destruction could be contained or reported through channels the daimyō controlled. Exit, by contrast, crossed domain boundaries by construction: fleeing villagers placed themselves under the protection of a rival lord or a shogunal intendant, who then had their own reasons for reporting the episode. A lord whose peasants fled therefore advertised his governance failure directly to rival lords and to the shogunate, making the same underlying attainder risk much more difficult to suppress (Steele, Paik, and Tanaka 2017). This asymmetry in publicity explains why daimyō had especially strong incentives to preempt exit by moderating extraction, particularly in border villages with credible access to outside authority. Consistent with this, mass flight was "what the warrior class feared most" (Kuromasa 1927, 69) and was prohibited by strict decree.

3.2.2. Exit as Deliberate Protest

Both exit routes, desertion and end-run appeals, depended on proximity to outside authority.[10] The two tactics were frequently deployed together in the same episode: village representatives would present petitions to external authorities while other members sought refuge in safe areas (Hosaka 2002; Miyazaki 1995).

Both served as bargaining tactics rather than permanent relocations. In a typical exit protest, some representatives traveled to file a petition with an outside authority while others took refuge in nearby foreign territory (Hosaka 2002; Miyazaki 1995). What exiters sought from outside officials was not immediate permanent resettlement: the outside authority could offer temporary shelter, guards, mediation, or formal recognition of a petition, and the possibility of a durable jurisdictional transfer mattered mainly because it strengthened bargaining leverage. Many episodes therefore ended not with permanent migration but with concessions from the home domain and a negotiated return (Aoki 1975; Miyazaki 1995). This bargaining-tactic character justifies the setup of my theoretical model, in which exit is a threat of departure resolved by concessions from the home ruler rather than a one-way relocation. Desertion and end-run appeals were, in short, deliberate protests, not passive flight (Hosaka 2002; Vlastos 1986).

End-run appeals in practice typically routed through neighboring rulers, who provided shelter, mediation, or escalation to the shogunate, rather than directly to Edo. Direct-to-Edo petitions, the Sakura Sōgorō pattern, are the well-known exception rather than the modal form (Walthall 1997). The three illustrative cases below all follow the neighbor-routed pattern: in Nobeoka 1690, Takanabe first mediated and then escalated to Edo; in Yamashirodani 1837, Imabari sheltered the villagers and offered absorption; in Matsuyama 1741, Ōzu forced concessions directly (Hosaka 2002; Miyazaki 1995). The

[10]Some cases coded as desertion in Aoki (1975) were passive exits known as "hashiri," where small groups fled secretly without political coordination or external recognition (Hosaka 2002). I retain these in the statistical analysis because they share the same strategic feature as deliberate chōsan: even passive flight required recognition from an external authority (without it, domain officials could track down fugitives and forcibly repatriate them), so success hinged on access to a neighboring jurisdiction.

distance to the nearest foreign capital d_i^{FC} is therefore the operative distance for both desertion and end-run appeals, since both depend on access to a rival ruler. Pooling the two tactics under "exit" is consistent with the theoretical mechanism: both depend on access to a rival ruler, not on access to the shogunate in the abstract.

3.2.3. How Exit Worked: Illustrative Cases

Three episodes from the secondary literature on *hyakushō ikki* (peasant uprising) illustrate how exit worked in practice: fleeing villagers secured shelter, mediation, and help escalating appeals to the shogunate from neighboring rulers, which made the threat of permanent relocation credible enough to extract concessions from the home lord.

The Nobeoka case (1690) illustrates how exit could coincide with the collapse of a ruling family. After years of oppressive taxation compounded by repeated flooding, peasants in Nobeoka Domain (Hyūga Province, southeastern Kyushu) fled south to the neighboring Takanabe Domain. Takanabe officials halted the fleeing villagers, offered to mediate, and — when negotiations reached an impasse — helped escalate the case to the shogunate in Edo, providing guards to escort village representatives and shelter for those who remained. Nobeoka acquiesced to the villagers' demands, including tax relief, and imposed no punishment. The following year, the shogunate stripped the Nobeoka lord Arima Kiyozumi of his domain and transferred him to the far smaller Itoigawa Domain in Echigo Province, effectively ending the Arima family's prominence (Ishikawa 1981).

The Yamashirodani case (1837) illustrates how neighboring lords actively facilitated exit, turning inter-domain competition into a credible threat. Villages in a mountainous district on the border between Tokushima Domain (Awa Province, Shikoku) and Iyo Province crossed into Imabari Domain. Rather than turning the peasants away, the Imabari lord dispatched guards to shield them from Tokushima's officials and facilitated negotiations, even warning that he was prepared to accept the villages into his own jurisdiction should Tokushima refuse their petition (Okamura 2009). The receiving domain had its own reasons to cooperate: absorbing productive villages meant more tax revenue

(Miyazaki 1995).[11]

The Matsuyama case (1741) shows exit operating as bargaining leverage rather than permanent relocation. After famine left them unable to pay taxes, several villages in Matsuyama Domain (Iyo Province, Shikoku) fled upriver to the neighboring domain of Ōzu and appealed for permission to migrate. Upon arriving in the Ōzu castle town, their strategy proved effective: the Matsuyama lord ultimately accepted most of their demands. Rather than punishing the revolt's leaders, the domain exiled its own chief minister to an island, attributing the crisis to his mismanagement (Nakagawa 1980). The villagers returned home.

The statistical analysis treats villages as strategic decision units choosing among compliance, fight, and exit based on distance, taxation, and collective-action technology. A substantial specialist literature (Vlastos (1986) on the political economy of benevolence, Walthall (1997) on ritualized mass petitioning, and White (1995) on the "referents of consciousness" invoked by protesters) emphasizes that peasants framed their actions in terms of customary legitimacy, honor, and ritual. The three illustrative cases above involve such elements: oaths, the role of the *otsukai* village representative, and the mino-and-spear symbolism of *gōso*. The strategic framing adopted here is appropriate as a first-cut model for testing aggregate statistical patterns across 910 revolts; it is not a claim about the complete phenomenology of Tokugawa protest. The moral-economy account and the strategic account are complementary: customary legitimacy shapes which actions are available in a village's repertoire, and relative costs among available actions shape which is selected.

3.3. Exposure Raises Exit Occurrence

Two designs test whether exposure to alternative authority raised the occurrence of exit.

[11]The Imabari–Tokushima episode is documented in Okamura (2009) and is consistent with the broader pattern of inter-domain labor competition documented for mountain border districts (Hosaka 2002; Miyazaki 1995).

3.3.1. Shock-Based Test: Tenryō Conversion

I leverage a quasi-experimental shock that introduced a powerful new competitor in close proximity: the Tokugawa shogunate's abolition of certain domains and the subsequent conversion of their territories into direct shogunal land (*tenryo*).

A tenryō conversion heightened governance competition for neighboring domains through several reinforcing channels. Tax burdens in tenryō were, on average, lower than in daimyō domains (Murakami 1996; Sng and Moriguchi 2014), with a few exceptions (Vlastos 1986). The installed intendants held short, rotating tenures and lacked independent military power (White 1995), reducing their investment in aggressive extraction. Contemporary peasants recognized and acted on these differences: White (1995) documents "a general perception that the administrative hand weighed lighter" in shogunal lands, such that calls for shogunal-style governance became "a component of peasant appeals," and records a significant number of incidents in which the people of an area demanded transfer to shogunal control (White 1995, 110). Moore (1993, p. 242) reaches the same conclusion in synthesis: "the hold of the authorities over the peasants was weaker in areas directly controlled by the Shōgun than in some of the outlying fiefs." A recent nearby conversion also served as precedent, signaling that the shogunate was willing to absorb additional territory. End-run appeals also became cheaper: with a shogunal intendant (*daikan*) newly installed at the former domain capital, nearby villagers could petition locally rather than undertake the costly journey to Edo.

I exploit these conversions to test Proposition 1's partial-equilibrium prediction, which captures the effect of a neighbor shock before the home ruler has optimally adjusted his own tax rate in response: a shift that raises y_{-j} and lowers τ_{-j} should raise exit but not fighting in foreign villages near the converted capital. Because conversions were sometimes triggered by internal disorder in the converted domain itself, the null on nearby fight doubles as a placebo that rules out a generic spillover of unrest from converted territories onto their neighbors.

I draw on the revolt data compiled by Aoki (1975), which catalogs 7,664 incidents of peasant resistance from local histories, official documents, and diaries.[12] Each observation records the date, home domain, nearest neighboring domain,[13] and mode of resistance classified as "fight" or "exit" following the categorization described in Section 3.2; I exclude 574 incidents classified as peaceful petitions and inter-village disputes (resource and boundary conflicts between villages rather than against the lord). I georeference each revolt village using several sources,[14] identifying the location of 2,571 villages out of the 2,790 sites recorded in Aoki (1975) the remaining sites are excluded due to imprecise location information. The resulting dataset contains 910 revolt incidents involving 2,571 villages and 2,741 tactic choices (summary statistics in Table A.5, Appendix B). As with most historical datasets, questions of completeness naturally arise; large and consequential mass exits are much more likely to have been preserved in the historical record than smaller-scale flights, so the absence of minor cases is unlikely to bias the core conclusions. I complement the revolt data with capital locations and creation and abolition dates for all 294 domains from Kimura, Fujino, and Murakami (2015) and Kodama and Kitajima (1989) (summary statistics in Table A.6, Appendix B); branches (shi-han) are treated as part of their main domains (hon-pan).

From these data I construct an annual domain-year panel covering 1615 to 1868. For each domain and year, I count exit and fight revolts occurring in villages ruled by other domains but located within a fixed radius of the domain capital. The main specification uses a $r \in \{30\text{km}, 40\text{km}, 50\text{km}\}$ radius around each capital to define the zone in which foreign villages are exposed to the new exit opportunity created by conversion. Figure 3 illustrates the design.

I exclude the four abolitions in which territory passed directly to another domain instead of to tenryo. The resulting 55 treated domains, enumerated with year and stated

[12] Hosaka (2002) characterizes Aoki (1975) as the culmination of postwar Japanese research on peasant uprisings and "one of the largest contributions in Japanese historical studies after WW2."

[13] I used Kodama and Kitajima (1989) to identify the neighboring domains existing at the time of each revolt. The nearest foreign domain is identified using time-varying domain existence data: for each revolt, only domains that were active at the time of the incident are considered as potential alternative authorities.

[14] These include Honda, Natsume, and Nemoto (2022), the geocoding API provided by the Ministry of Land, Infrastructure, Transport and Tourism of Japan, and the Historical Location Name Comprehensive Database provided by the Center for Open Data in the Humanities.

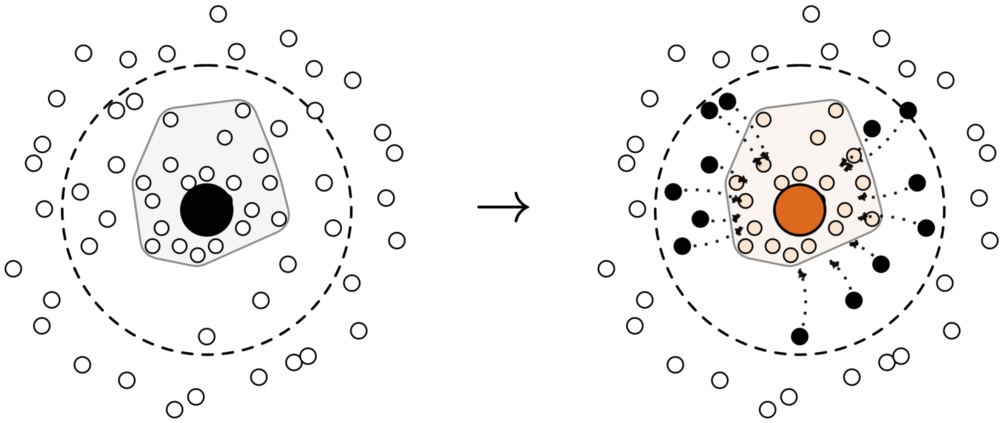


Figure 3: DiD Design: Tenryō Conversion

Note: Stylized illustration. Left: a daimyō domain (gray territory) extracts taxes; foreign villages within radius r of the capital (solid black dots outside the territory) face no attractive alternative authority. Right: abolition converts the same capital into shogunal territory (*tenryō*, orange), which was on average less oppressive; those same foreign villages can now exit toward the converted territory (dotted arrows). The DiD compares changes in exit and fight outcomes in these treated foreign villages against never-converted neighbors.

reason in Appendix C.1, are drawn from 79 dissolutions recorded across the 294-domain universe in Kimura, Fujino, and Murakami (2015) and Kodama and Kitajima (1989). For converted domains, I then restrict the sample to a symmetric event window from five years before to five years after the first tenryo-conversion year, keeping only strict pre-conversion years and years in which the domain is actually dissolved under direct shogunal rule. Never-converted controls are kept only in the same calendar years represented by the converted observations in each subset. I estimate annual domain-level linear probability difference-in-differences with domain and year fixed effects:

$$\text{Revolt}_{j,t} = \beta_1\big(\text{Converted}_j \times \text{Post}_{j,t}\big) + \varphi_j + \omega_t + \varepsilon_{j,t}$$

In this specification, $\text{Revolt}_{j,t} \in \big\{\text{Exit}_{j,t}, \text{Fight}_{j,t}\big\}$ represents a binary indicator for the occurrence of any nearby exit or fight within r kilometers of domain j's capital in year t. Converted_j denotes domains that ever undergo a tenryo conversion within the sample, while $\text{Post}_{j,t}$ marks years following the first such conversion when the domain is under direct shogunal administration.

A caveat is that the destination of each recorded exit is largely unknown: the sources note where fleeing villagers came from but rarely how and where they traveled. Proximity to the converted capital therefore proxies for the converted territory being a plausible destination rather than directly measuring it. Two features of the results are reassuring. First, tightening the radius from 50km to 40km and 30km preserves the sign and significance of the exit coefficient on progressively more spatially precise exposure bands. Second, the placebo null on nearby fight holds at every radius, which is difficult to reconcile with a story in which the exit coefficient merely picks up geographically mismeasured unrest.

Table 1 shows a positive nearby-exit response. In the main 50km specification, tenryo conversion raises the probability of any nearby exit by 0.0385 ($p = 0.010$), about 3.9 percentage points, or roughly 59% of the control-group pre-conversion mean (0.065), while the corresponding any-fight estimate is only 0.0076 ($p = 0.532$). When a nearby domain was replaced by direct shogunal rule, surrounding villages became more likely to run than to stay and fight.

Outcome:	Any Nearby Exit			Any Nearby Fight		
Sample Radius:	30km	40km	50km	30km	40km	50km
Variables						
Converted × Post	0.0163	0.0262	0.0385	0.0084	0.0033	0.0076
Cluster SE	(0.0078)**	(0.0116)**	(0.0147)***	(0.0093)	(0.0092)	(0.0122)
Conley SE (70km)	(0.0084)*	(0.0107)**	(0.0131)***	(0.0104)	(0.0108)	(0.0136)
Controls						
Domain FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Fit Statistics						
Num. Obs.	33,877	33,877	33,877	33,877	33,877	33,877
Tenryo-Converted/All Domains	55/294	55/294	55/294	55/294	55/294	55/294
R2	0.038	0.052	0.062	0.078	0.101	0.136
Mean Dep. Var.	0.010	0.018	0.026	0.020	0.031	0.044

Table 1: Domain-to-Tenryo Conversions Increase Nearby Exit

Note: LPM estimates on an annual domain-year panel. “Nearby” denotes revolts by villages of *other* domains within the specified radius. Converted domains restricted to rel_year in [−5, +5], matched to never-converted controls in the same calendar years. SEs in parentheses: (i) two-way clustered (domain, year); (ii) Conley (70 km) on capital coordinates; stars reflect that estimator’s p-value. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

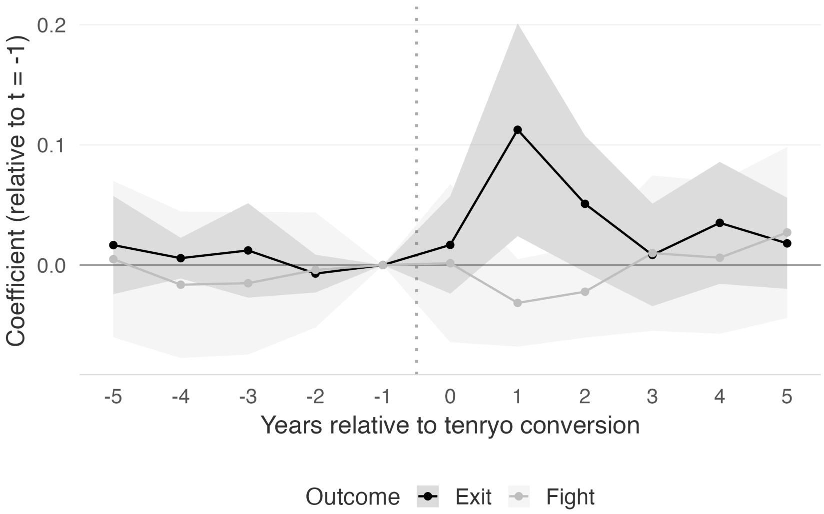


Figure 4: Tenryo-Conversion Event-Study Coefficients

Note: Year-by-year event-study coefficients with $s = -1$ as the reference period for both nearby exit (top) and nearby fight (bottom). Shaded bands show 95% confidence intervals. "Nearby" denotes revolts by villages belonging to *other* domains but located within 50km of the focal domain's capital. Two-way clustered (domain & year) standard errors.

The identification assumption is that converted and never-converted domains would have followed parallel revolt trends absent conversion. I assess this with an event study:

$$\text{Revolt}_{j,t} = \sum_{s \neq -1} \beta_s \big(\text{Converted}_j \times \mathbf{1}\left[t - t_j^* = s\right]\big) + \varphi_j + \omega_t + \varepsilon_{j,t}$$

where Converted_j denotes tenryo-converted domains, t_j^* is the first conversion year, and $s = -1$ is the reference period. As shown in Figure 4, pre-treatment coefficients are individually small and a joint F-test confirms their insignificance ($F = 1.26$, $p = 0.283$; fight: $F = 0.18$, $p = 0.947$). After conversion, the exit series jumps sharply at $t = 1$ ($\hat{\beta}_1 = 0.113$, $p < 0.05$) and remains elevated throughout the window, while the fight series stays flat. A Rambachan and Roth (2023) relative-magnitudes sensitivity analysis yields a breakdown value of $\overline{M}^* = 0.20$: the exit effect tolerates post-period deviations up to 20% of the worst pre-period deviation (Appendix C.7). At canonical benchmarks $\overline{M} \in \{0.5, 1, 2\}$ and under the absolute-smoothness restriction Δ^{SD} evaluated at the pre-period standard deviation σ_{pre}, the signed effect is preserved though confidence intervals widen (Appendix C.8). A complementary identification check models the timing of tenryo conversion directly: lagged nearby exit and fight intensities do not predict which domain the shogunate converts in a given year (joint Wald $p = 0.344$, Appendix C.11).

The results are robust to alternative specifications and estimation strategies. Because the 55 conversions are staggered across more than two centuries, the standard TWFE estimator could be biased if treatment effects vary across cohorts (Baker, Larcker, and Wang 2022). A Goodman-Bacon decomposition places 94.5% of the TWFE estimand weight on clean treated-versus-never-treated comparisons, leaving the "bad-comparison" problem of heterogeneous-timing DiD immaterial here (Appendix C.5). Consistently, the interaction-weighted estimator of Sun and Abraham (2021), which is robust to staggered-timing heterogeneity, yields an aggregate exit ATT of 0.0413 ($p < 0.001$), virtually identical to the TWFE estimate of 0.0385, while the fight ATT remains insignificant (0.0089, $p = 0.262$); the full comparison appears in Appendix C.4. A cohort-split Sun-Abraham estimate concentrates the exit effect in the pre-1750 cohort, which holds 49 of 55 conversions (ATT $= 0.044$, $p < 0.001$), with the two later cohorts too small to be load-bearing (Appendix C.6).

The tighter 30km and 40km bands confirm the same pattern, with exit coefficients significant at 0.0262 ($p = 0.025$) and 0.0163 ($p = 0.038$) and fight coefficients indistinguishable from zero across all radii. Conley standard errors at bandwidths from 30 to 150 km deliver p-values between 0.0002 and 0.0033 for the 50 km exit estimate; the Kelly (2019) data-driven bandwidth selects a 10 km cutoff that produces inference within 0.001 of the hand-picked 70 km bandwidth (Appendix C.12, Appendix C.13). Replacing the binary outcome with exit and fight counts produces consistent estimates (Appendix C.2). A 500-iteration permutation test that randomly reassigns fake conversion years to never-converted domains yields a two-sided p-value of 0.012 for the actual estimate; a more demanding variant that instead draws 55 placebo conversions from the full pool of 294 domains (including the actually-converted 55) yields $p = 0.006$ (Appendix C.10). Both variants confirm the response is specific to the actual timing and location of conversions, not an artifact of the panel's staggered-adoption structure. Because some conversions cluster spatially, I also verify that the result is not driven by interference among overlapping treatment radii: dropping all converted domains within 100 km and 5 years of another conversion, or excluding controls within 50 km of any converted capital, preserves the sign and magnitude of the exit coefficient (Appendix C.9). The result also holds when exit is restricted to desertion alone, a narrower definition that excludes end-run appeals (Appendix C.3).

The effect loads on the regime that best fits the mechanism. The cohort split (Appendix C.6) shows the exit response concentrated in the pre-1750 cohort, which holds 49 of 55 conversions. This is consistent with the specialist view that the attainder threat, the primary publicizing risk created by exit, was most credible in the first century of Tokugawa rule, when the shogunate actively disciplined daimyō (Ravina 1999; White 1995). Later conversions, particularly after the Tenmei and Tenpō reforms of the 1780s and 1840s, operated in a different political environment: weakened shogunal authority, rising commercial taxation, and broader fiscal strain. The headline effect therefore loads primarily on the regime in which the attainder-publicity mechanism had the most bite.

3.3.2. General Test: Grid-Year Panel

The DiD identifies exit's response to a specific shock. A complementary general test asks whether geographic exposure in general triggers exit, as predicted by Proposition 2(b). A village-level panel would be natural, but domain affiliation is observed only at the end of the Tokugawa period — for most non-revolting villages, the governing domain at any given date is unknown — making a village panel with time-varying distance ratios impossible. A grid-year panel sidesteps this by assigning villages to fixed 5km hexagons and computing the ratio from the two nearest active capitals each year, without requiring domain affiliation. The outcome counts exits per hexagon-year.

I construct a 5km grid-by-year panel that recomputes the distance ratio every calendar year using the two closest active capitals, so each cell-year treatment reflects domain dissolutions, tenryo transfers, or reestablishments, not a single event. Populated cells are defined from the 46,086-village LFAT catalog introduced in Section 3.4: each village is assigned to the 5km hexagon containing it. The panel retains 2,225,802 cell-years across 8,763 occupied cells, and each observation counts the number of exit and fight revolts within that hexagon in the given year.

I introduce the paper's central predictor here: the *distance ratio*, a location-level measure of exposure to jurisdictional conflict. At the grid level,

$$\tilde{\theta}_{i,t} := \frac{d^1_{i,t}}{d^2_{i,t}}$$

where $d^1_{i,t}$ and $d^2_{i,t}$ are the straight-line distances from hexagon i's centroid to its nearest and second-nearest active domain capitals in year t. The ratio rises whenever a new capital appears closer than the existing second-nearest (as when a neighboring domain is dissolved or a new one established), capturing the relative accessibility of alternative authority, which proxies the intensity of jurisdictional conflict the cell faces. A village-level analog (Section 3.4.1) is used in the next section.

The estimating equation is

$$\text{Revolt}_{i,t} = \beta_3 \tilde{\theta}_{i,t} + \lambda_i + \eta_t + \varepsilon_{i,t}$$

with hexagon and year fixed effects. Here $\text{Revolt}_{i,t}$ is either the exit count or the fight count. The distance ratio is recomputed each year as described above, so identifying variation comes from capital dissolutions and transfers and the resulting reshuffling of the nearest outside authority. Figure 5 illustrates this variation by mapping the distance ratio across three snapshot years. The overall pattern is stable, but localized changes are visible: in areas where new domains were established or existing ones dissolved, competition intensity shifts as hexagons gain or lose proximity to the closest or the second closest capital. Hexagon and year fixed effects absorb time-invariant geography and common shocks.

Table 2 shows that a higher distance ratio predicts exit incidence but not fight incidence, and that mere remoteness from any state authority predicts neither. Columns (1)–(3) report LPM estimates with the time-varying distance ratio: a one-unit increase is associated with a 0.035-percentage-point higher probability of any exit in a cell-year, statistically significant at the 5% level under both two-way cell-and-year clustering ($p = 0.048$) and Conley spatial inference ($p = 0.037$). The corresponding fight coefficient is smaller, positive, and not statistically significant. The exit coefficient is small in absolute terms — the baseline probability of any exit in a cell-year is only 0.022% — but large

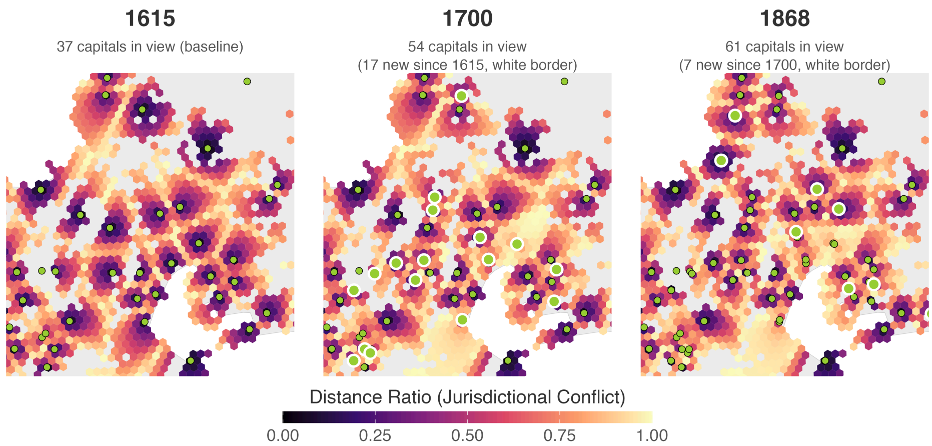


Figure 5: Time-Varying Distance Ratio across 5km Hexagonal Grid: Kinki Region

Note: Each panel maps $\tilde{\theta}_{i,t}$ (distance to nearest active capital / distance to second-nearest) for a given year, zoomed to a $\approx 200\,\text{km} \times 200\,\text{km}$ window in central Japan (lat 34.4–36.2°N, lon 135.5–137.6°E) covering Kyoto, Osaka, Nagoya, and Lake Biwa — the region with the highest concentration of cells whose distance ratio changed across the three snapshot years. Yellowgreen circles mark active capitals; white borders distinguish capitals newly added since the previous snapshot. Higher values (yellow) indicate strong jurisdictional competition; lower values (purple), a dominant nearest capital. Spatial shifts reflect domain dissolutions and creations.

	Distance ratio ($\tilde{\theta}_{i,t}$)			Closest-capital dist. ($d^1_{i,t}$, km)		
Outcome:	(1)	(2)	(3)	(4)	(5)	(6)
	Any exit	Any fight	Any fight (strict)	Any exit	Any fight	Any fight (strict)
Coefficient	0.00035	0.00026	−0.00012	0.00000	0.00000	0.00000
Cluster SE	(0.00017)**	(0.00019)	(0.00011)	(0.00000)*	(0.00000)	(0.00000)
Conley SE (10km)	(0.00017)**	(0.00023)	(0.00012)	(0.00000)	(0.00000)	(0.00000)
Controls						
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Fit Statistics						
Num.Obs.	2,225,802	2,225,802	2,225,802	2,225,802	2,225,802	2,225,802
R^2	0.006	0.007	0.006	0.006	0.007	0.006
Mean Dep. Var.	0.00022	0.00064	0.00026	0.00022	0.00064	0.00026

Table 2: Jurisdictional Competition Predicts Exit

Note: LPM on binary any-event-of-type indicators at the 5 km hexagon–year level for the full panel of 8,763 cells. Columns (1)–(3) use the time-varying distance ratio $\tilde{\theta}_{i,t} = \frac{d^1_{i,t}}{d^2_{i,t}}$ (computed from the two closest active capitals each year). Columns (4)–(6) replace it with the raw distance to the closest active capital $d^1_{i,t}$ (km), a measure of mere remoteness from any state authority that discards information about the second-nearest. “Fight (strict)” restricts fight to destruction and uprising. Grid and year FE. SEs: (i) two-way clustered (cell, year); (ii) Conley (10 km). The ever-active-cells subsample analysis appears in Appendix D.2.12. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

relative to this base: a one-standard-deviation increase in the distance ratio (SD = 0.26) corresponds to roughly 40% of the baseline mean.

Columns (4)–(6) replace the distance ratio with the raw distance to the closest active capital $d^1_{i,t}$, a measure of mere remoteness from any state authority that discards information about the relative position of the second-nearest. The closest-capital coefficient on exit is small and only marginal under clustered SE ($p = 0.092$) and gets null under Conley SE ($p = 0.162$); the fight and fight-strict coefficients are likewise indistinguishable from zero. The contrast between columns (1) and (4) isolates the active ingredient: the relative geometry of competing rulers, not the absolute distance to the nearest one, is estimated precisely enough to predict exit. Restricting the panel to ever-active cells (the 1,466 hexagons with at least one event between 1615 and 1868) preserves the result while statistical significance attenuates somewhat (Appendix D.2.12). The result is robust to count-data estimators (Poisson and Negative Binomial FE; Appendix D.2.11), restricting exit to desertion only (Appendix D.2.10), replacing great-circle distance with a terrain-weighted least-cost-path ratio (Appendix D.2.13), and including daikansho in the distance computation (Appendix D.2.15), though statistical significance attenuates somewhat in each.

3.4. Exposure Shifts Revolt Composition Toward Exit

The two preceding tests establish that exposure raises exit occurrence. A distinct question is whether, among villages that did revolt, greater geographic exposure shifted the chosen form of resistance toward exit over direct confrontation, as predicted by Proposition 2(a).

The village-level analyses here and below (tax rates in Section 3.5) draw on a dataset of 46,086 villages compiled primarily from the *Kyūdaka–Kyūryō Torishirabe Chō* (Ledgers of Former Assessments and Territories, hereafter LFAT), a comprehensive Meiji-government land census that records the home domain for each village at the end of the Tokugawa period; village location data are obtained from Honda, Natsume, and Nemoto (2022). When multiple domains collected taxes from a single village, I assign it to the domain possessing the largest share of its assessed productivity (*kokudaka*, a rice-denominated index of aggregate agricultural output rather than literal rice harvest alone). The resulting data encompass virtually all of Japan's arable land, omitting only high-altitude, non-cultivable regions, as illustrated in Figure 2(a)–(b) and Figure 7.

3.4.1. Measuring Exposure to Alternative Authority

The theoretical parameter θ_j is a domain-level binary share: the fraction of villages facing low exit costs. I operationalize this at the village level as a continuous index $\hat{\theta}_i$, measuring each village's geographic exposure to alternative authority. The logic is that villages closer to an outside authority face lower exit costs because proximity facilitates the external intervention on which mass exit depended (Section 3.2): it reduced the cost of dispatching guards or mediators to protect fleeing peasants, made it easier for villagers to reach neighboring capitals or shogunal offices to voice grievances, and made administrative transfer of allegiance more practicable. I therefore construct a parsimonious distance-based measure:

$$\hat{\theta}_i := \frac{d_i^{\mathrm{HC}}}{d_i^{\mathrm{FC}}}$$

where d_i^{HC} denotes the straight-line (Euclidean) distance from village i to its home-domain capital, and d_i^{FC} is the straight-line distance to the nearest non-home domain capital that was active at the time of the revolt.[15] Both distances are calculated using georeferenced village locations from the revolt data and the time-varying domain-capital panel described in Section 3.3.1. A capital is considered "active" in a given year if it served as the seat of an existing daimyō domain, or if the domain had been dissolved and converted to tenryo, in which case the daimyō was replaced by a shogunal intendant (*daikan*) (Vlastos 1986, 49). In the latter scenario, the capital remains in the choice set for d_i^{FC} throughout the tenryo period.[16]

[15] Robustness checks employing least-cost path distances that incorporate terrain are reported in Appendix D.2.4.

[16] In some cases, the shogunate may have administered converted territory from a nearby *daikansho* (a shogunal intendant's district office) rather than from the former castle town. To address this, Appendix D.2.14, Appendix D.2.15, and Appendix D.3.1 replicate all main analyses with the nearest capitals redefined to include *daikansho* locations. Nevertheless, Aoki's destination data indicate that daikansho seldom served as actual exit destinations (none appear as targets), and, in line with this, all replications yield attenuated results.

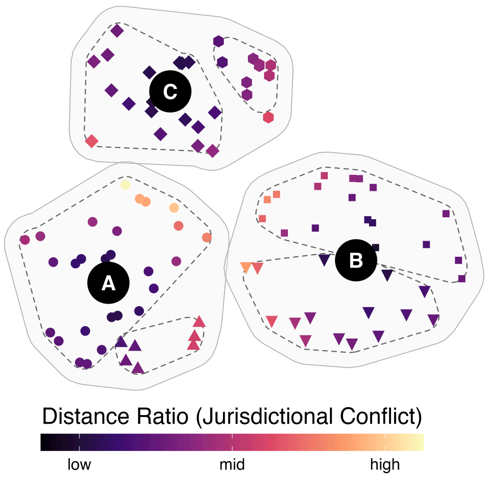


Figure 6: Dyad FEs Enable Within-Pair Comparison

Note: Three stylized domains (A, B, C) with capitals (black circles) and villages (shapes) shaded by the distance ratio ($\hat{\theta}_i$): dark shades indicate low exposure (villages near their home capital), light shades indicate high exposure (villages near a rival capital). Home-neighbor dyad fixed effects restrict comparisons to villages sharing both home and nearest-rival domains (those within the same dashed circles).

3.4.2. Empirical Strategy: Home-Neighbor Dyad Fixed Effects

Each observation corresponds to a village involved in a revolt recorded in Aoki (1975).

To isolate variation in the distance-ratio measure from potential confounders, I employ home-neighbor dyad fixed effects. Each dyad denotes a unique ordered pair of home domain and nearest-rival domain (e.g., A–B, A–C in Figure 6). Because dyad FEs nest both home-domain and neighbor-domain FEs, they absorb every unobserved domain-level feature — each domain's tax rate, coercive capacity, ruler ideology, agricultural endowment, and administrative quality — alongside pair-specific factors such as historical cross-border ties, shared transit corridors, or pair-level border-control arrangements. Identification then comes from villages that share both home and nearest-rival domains but differ in their relative distance to the two capitals — the within-dyad variation Figure 6 depicts inside each dashed circle.

In sum, the main specification is:

$$\text{Exit}_{i(v,j,k,t)} = \beta_2 \ln\left(\hat{\theta}_i\right) + X_v' \boldsymbol{\gamma_2} + \underbrace{\varphi_{j,k}}_{\text{home-neighbor dyad FE}} + \underbrace{\omega_t}_{\text{decade FE}} + \varepsilon_i$$

Here, Exit_i is a binary variable indicating whether event i is exit (1) or fight (0). The main predictor, $\hat{\theta}_i$, represents the ratio of the distance from village v to its home domain capital j relative to the distance to the nearest non-home domain capital active at the time of event i, as defined in the preceding subsection. This ratio is log-transformed to stabilize variance.[17] The vector X_v contains village-level controls: latitude, longitude, their interaction, elevation (log), ruggedness (log), distance to the nearer of Edo or Kyoto (where the shogunate's headquarters was located, log), and wet rice suitability (log).[18] The home-neighbor dyad fixed effects ($\varphi_{j,k}$) absorb all unobservables specific to each ordered pair of home and nearest-rival domains. Decade fixed effects (ω_t) account for temporal factors that might influence rebellion patterns throughout the study period.

3.4.3. Geographic Exposure Predicts Exit Share

The estimation results in Table 3 show that the distance ratio measure predicts villages' propensity to opt for exit over fight when rebelling. The coefficients remain positive across all specifications and statistically significant in the preferred controlled specifications, indicating that villagers strategically select their mode of resistance based on their relative distances to home and outside authorities. In the preferred specification with home-neighbor dyad fixed effects and full controls (Model 5), a one-standard-deviation increase in the log distance ratio (≈ 1.58) raises the exit probability by 3.9 percentage points, or about 16% of the 24.6% baseline exit rate. The less restrictive two-way (home + neighbor) FE specification (Model 4) yields a slightly smaller but statistically indistinguishable estimate of 3.4 percentage points, indicating that the headline result does not hinge on the dyadic restriction.

Decomposing the ratio into its two absolute distances reveals that proximity to a *foreign* capital is the active margin on two counts. First, the foreign-capital coefficient exceeds the home-capital coefficient in absolute magnitude: in the preferred dyadic speci-

[17] Using the logged ratio of distances requires caution, as it effectively constrains the coefficients of the two logged distance variables to be equal in magnitude but opposite in sign. Reassuringly, linear hypothesis tests confirm that this restriction is consistent with the data, showing no evidence that the coefficients for the two distances differ statistically from one another, as reported in Appendix D.3.5.

[18] Wet rice suitability is measured as potential output under irrigation and medium input, based on the Caloric Suitability Index developed by Galor and Özak (2016).

Outcome:				Exit_i			
			Distance Ratio			*Abs. Distances*	
Models:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variables							
Distance ratio ($\ln(\hat{\theta}_i)$)	0.0458	0.0519	0.0201	0.0212	0.0244		
Cluster SE	(0.0154)***	(0.0141)***	(0.0071)***	(0.0072)***	(0.0068)***		
Conley SE (10km)	(0.0117)***	(0.0093)***	(0.0078)***	(0.0078)***	(0.0085)***		
$\ln(d_i^{\text{HC}})$						0.0160	0.0203
Cluster SE						(0.0104)	(0.0083)**
Conley SE (10km)						(0.0106)	(0.0113)*
$\ln(d_i^{\text{FC}})$						−0.0306	−0.0309
Cluster SE						(0.0129)**	(0.0135)**
Conley SE (10km)						(0.0138)**	(0.0148)**
Controls							
Decade FE	No	Yes	Yes	Yes	Yes	Yes	Yes
Home Domain FE	No	No	Yes	Yes	No	Yes	No
Neighbor Domain FE	No	No	Yes	Yes	No	Yes	No
Home-Nbr Dyad FE	No	No	No	No	Yes	No	Yes
Geography	No	No	No	Yes	Yes	Yes	Yes
Fit Statistics							
Num.Obs.	2,739	2,739	2,739	2,739	2,739	2,739	2,739
R2	0.028	0.225	0.621	0.626	0.678	0.627	0.678
Mean Dep. Var.	0.246	0.246	0.246	0.246	0.246	0.246	0.246

Table 3: Exposure to Alternative Authority Predicts Exit (Exit versus Fight)

Note: OLS estimates of each village's exit-vs-fight choice conditional on revolting. Models (1)–(5) use the log distance ratio; Models (6)–(7) replace it with the two absolute distances (d_i^{HC}: home capital; d_i^{FC}: nearest foreign capital). Full results in Table A.22. SEs in parentheses: (i) clustered on home, nearest-neighbor, decade; (ii) Conley (10 km) on village coordinates; stars reflect that estimator's p-value. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

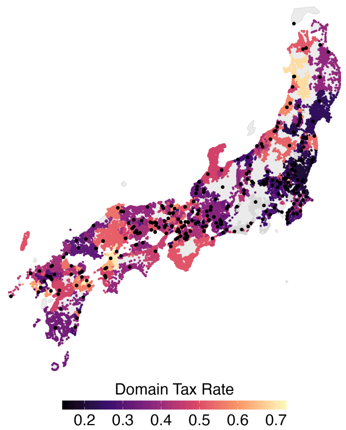
fication (Model 7), a one-standard-deviation decrease in $\ln(d_i^{\text{FC}})$ (≈ 0.92) raises the exit probability by 2.8 percentage points, or about 11% of the 24.6% baseline exit rate, with a near-identical magnitude under two-way FE (Model 6). Second, in Model 7, $\ln(d_i^{\text{HC}})$ loses statistical significance once Conley spatial standard errors are imposed, whereas $\ln(d_i^{\text{FC}})$ remains statistically significant under both cluster and Conley standard errors in every specification. Access to a rival center of power, not limited oversight by the home domain in isolation, drives villages toward exit.

The result is robust across alternative specifications, controls, and measurement strategies. The coefficients hold under alternative controls for agricultural productivity and distance to political centers (Appendix D.2.1) and after excluding repeat-rebel villages (Appendix D.2.2). Using least-cost path distances that account for terrain ruggedness produces results consistent with the main analysis, though less robust across specifications (Appendix D.2.4); fleeing villagers likely took circuitous routes to avoid detection rather than optimal paths, weakening the relevance of the theoretically "least-costly" distance. A related concern is that straight-line distance can cut across sea straits, implying walking routes that are physically impossible; dropping village-neighbor dyads whose anchors lie on different islands leaves the coefficient unchanged (Appendix D.2.5). Controlling for each village's minimum distance to the nearest domain border (Appendix D.3.2), the distance ratio continues to predict exit while border distance itself is not statistically associated with exit choice, confirming that access to a rival center of power, not simply location near a boundary, drives the result. Restricting exit to desertion alone preserves the result (Appendix D.2.9). Including shogunal intendant offices (*daikansho*) in the distance computation leaves the preferred dyadic-FE estimate virtually unchanged (Appendix D.2.14). Finally, because Table 3 conditions on revolt, geographic exposure could in principle shape selection into the rebel set itself; a nested-logit decomposition that merges the rebel sample with the comprehensive end-period village ledger and estimates the revolt-versus-comply margin separately from the exit-versus-fight contrast finds a clean null at the selection step and recovers the Table 3 Model 5 coefficient at the form-of-resistance step, so the conditioning here is not a source of selection bias on $\hat{\theta}_i$ (Appendix D.3.3).

3.5. Exposure and Constrained Taxation

The preceding analyses establish that access to alternative authority channels resistance toward exit. The model predicts a further consequence: where exit is viable, rulers lower taxes to retain their population (Proposition 3). Testing this prediction directly is difficult because systematic tax data survive only as a late-period cross-section; a pooled regression of tax rates on geographic exposure is therefore associational and on its own indistinguishable from mechanisms such as state-capacity decay. I instead exploit variation in neighbor characteristics that shifts the intensity of jurisdictional conflict within the cross-section. The exit mechanism predicts that the negative association between geographic exposure and tax rates should be concentrated in domains whose neighbors are attractive enough to credibly absorb migrants; state-capacity decay implies no such asymmetry. To operationalize the geographic-exposure component, I calculate

(a) Village Log Distance Ratio (Jurisdictional Conflict)



(b) Domain Tax Rates

Figure 7: Geographic Exposure and Tax Rates Across Villages

Note: Panel (a) plots village locations shaded by the log distance ratio $\ln\left(\hat{\theta}_i\right) = \ln\left(d_i^{\mathrm{HC}}/d_i^{\mathrm{FC}}\right)$, the village-level measure of geographic exposure to alternative authority. Panel (b) plots the same village locations shaded by their domain's nominal tax rate; gray points denote villages in domains for which tax data are unavailable. Black dots mark domain capitals.

the average distance ratio across all villages within each domain. Figure 7 previews the two key variables: villages shaded by their log distance ratio (panel a) and by their domain's nominal tax rate (panel b).

3.5.1. Empirical Strategy

Estimating this association raises two identification concerns. The first is endogeneity of capital location: domain rulers may have strategically located their capitals or exchanged border villages to reduce the risk of mass exit. Capital locations were largely inherited from the Sengoku period (Hall 1955) and frozen by the 1615 one-castle-per-domain edict (*ikkoku ichijō rei*), with subsequent relocations requiring shogunal approval and remaining rare. Deliberate anti-exit capital placement after 1615 was therefore uncommon, limiting the scope for the endogeneity channel. Shogunate-strategic placement of fudai versus tozama daimyō (the shogunate seated hereditary vassals in strategically important positions and kept tozama lords on the periphery (Ravina 1999)) is a separate source of confounding that is partially absorbed by the alliance-status control below. Even if residual adjustments occurred, they bias results conservatively: deliberate capital placement and village transfers would attenuate, not inflate, the observed negative relationship between geographic exposure and tax rates. A separate concern is that a daimyō's ability to relocate a capital or trade border villages may have depended on political ties to the shogunate, since such changes required shogunal approval. This could confound the estimates if those ties also shaped a daimyō's extractive capacity. To address this possibility, I control for each daimyō's alliance status with the shogunate.[19]

The second concern is confounding. The home-capital distance component could proxy for domain scale, state capacity at the periphery (Sng and Moriguchi 2014), or village development stage, all of which might independently affect tax rates. I address these channels with controls for domain population,[20] agricultural productivity, territory size, the *gōshi system* (where lower-ranking samurai resided in villages as local officials, raising monitoring costs), and the proportion of villages with "shinden" (new rice field) in their names, indicating recent development. Tax delivery costs are unlikely confounders because villagers delivered taxes to nearby local storage facilities, not to domain capitals.[21] Tokugawa institutional arrangements also limited local agents' authority (see Section 3.1), mitigating the state-capacity-decay channel. Any remaining confounding through domain scale or coercive capacity should attenuate the estimate, since larger domains could maintain larger armies to enforce higher rates.

The outcome is the domain-level effective tax rate, computed from the same Kodama and Kitajima (1989) domain compendium introduced above (pp. 425–431) as the ratio of *shunodaka* (actual reported tax collections in rice equivalents) to *uchidaka* (the domain's internal record of actual rice production, distinct from the publicly declared *kokudaka* assessment). The ratio therefore captures the share of realized output that the domain actually collected, which is what the model's τ_j represents. The figures are based on statistics compiled by the Meiji government in 1869–1870, when each domain reported average production and collections for the preceding five years. The use of a multi-year average mitigates short-run harvest noise, and the widespread adoption of the *jōmen-*

[19]Daimyō are classified according to their relationship with the shogunate: shinpan (relatives), fudai (hereditary vassals who fought for the shogunate at Sekigahara in 1600), and tozama (outside lords who became allies only after the battle).

[20]Population is measured at the end of the Tokugawa period, so variables that use it should be read as late-period proxies rather than time-invariant characteristics.

[21]For instance, Fukui Prefecture (1994: 335-346) documents this practice in Fukui Domain.

hō (fixed-rate method) meant that many domains operated with administratively stable rates even in the final decades of Tokugawa rule (Iinuma 1974, 20–24; Smith 1958).

The late-period measurement window deserves comment. Both *shunodaka* and *uchidaka* are drawn from Meiji-era reports covering 1865–69, so the cross-sectional pattern they document is a late-period steady state rather than a time-varying outcome. Two features of the Tokugawa fiscal system make this measurement window informative for the underlying mechanism. First, *jōmen-hō* locked nominal rates in place across decades (Smith 1958), so within-domain year-to-year variation in effective rates was limited. Second, because the mechanism operates on long-run extractive stances rather than short-run fiscal shocks, a late-period measurement is appropriate for an equilibrium comparative-statics interpretation of Proposition 3. Mid-Tokugawa tax scatters compiled by Smith (1958) are qualitatively consistent with the late-period cross-section, though comprehensive mid-period data at the domain level are unavailable. Although tax rates varied among villages within each domain, systematic data at the village level are also unavailable; domain-level averages capture the main variation relevant for the analysis.[22]

I exclude non-daimyō territories (such as *goryō* under the shogunate's direct control and *jisharyō* administered by shrines and temples) as outcome units because these territories lack comprehensive fiscal data and possessed limited autonomy in establishing tax policies (Murakami 1996). The domain-level average $\hat{\theta}_j$ is computed from the same distance-ratio measure defined in Section 3.4.1, averaged across all villages within each domain using end-period capital locations.

3.5.2. The Tax–Exposure Link Binds Where Neighbors Attract Labor

I begin with a pooled benchmark that regresses the domain tax rate on geographic exposure and controls:

$$\tau_j = \alpha_4 + \beta_4 \ln\left(\hat{\theta}_j\right) + X_j'\boldsymbol{\gamma_4} + \varepsilon_j$$

[22]One might wonder whether public goods provision, rather than tax reduction, served as an alternative means for rulers to address peasant grievances. If so, tax rates could be an incomplete measure of oppressiveness. This is unlikely for two reasons. First, public works projects depended on corvée labor (bueki) extracted from peasant villages, which increased rather than alleviated their burdens. Second, evidence from peasant petitions and revolt records indicates that grievances centered primarily on excessive taxes and unfair assessments, not on the lack of infrastructure (Aoki 1975).

Here τ_j is the tax rate in domain j and $\hat{\theta}_j$ is the domain-average distance ratio; I log-transform the distance ratio to address skewness. The vector X_j contains three groups of controls: (1) Core: domain size (log), wet-rice suitability (log),[23] proportion of new-farmland villages, and indicators for shogunate allies and the gōshi system; (2) Geography: elevation (log), ruggedness (log), and average proportions of rainy and snowy days in a month;[24] and (3) Neighbors: weighted means of adjacent domains' characteristics (territory size, population, rice productivity, number of new farmlands, family ties with the focal domain, and shogunate alliance status), with weights based on the number of villages for which each domain's capital is the nearest neighboring capital.

Table 4 documents a negative cross-sectional association between geographic exposure and reported tax rates. A one-standard-deviation increase in the log distance ratio (SD = 1.34) corresponds to a 1.5–3.2 percentage-point decrease in the reported tax rate (Models 1–4), an association that persists with the full set of controls, in a pure spatial-lag model, and marginally in the fully controlled spatial-lag model (Anselin and Rey 2014) (Models 5–6). The direction aligns with Steele, Paik, and Tanaka (2017), who establish at the domain-period level that collective desertions, but not the broader insurrection category, are associated with lower tax rates, the aggregate correlate of the mechanism this paper investigates at the community level. The pooled estimate is robust to alternative covariate specifications (kokudaka, samurai share, per-capita productivity; Appendix E.2.1), a Diegert, Masten, and Poirier (2022) sensitivity analysis (an unobserved confounder would need to exceed half the strength of all included controls to flip the sign; Appendix E.2.2), double/debiased machine learning (Chernozhukov et al. 2018) over $p = 38$ candidate covariates ($\hat{\beta} \in [-0.012, -0.010]$, $p < 0.05$; Appendix E.2.3), and including daikansho in the distance measure (Appendix D.3.1). Because this cross-sectional estimate cannot on its own separate exit from state-capacity decay, the analysis below exploits neighbor heterogeneity to isolate the exit channel.

[23]The wet-rice measure is an agro-climatic potential-yield surface (Galor and Özak 2016) rather than an observed annual output series; I use it only as a proxy for labor-absorption capacity.

[24]Weather variables represent proportion of rainy/snowy days per month, imputed as weighted averages using daily weather data from 40 locations, with distance to domain capital as weight. Data from Historical Weather Database by Professor Minoru Yoshimura.

Outcome:	τ_j					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Distance ratio ($\ln(\hat{\theta}_j)$)	-0.0241	-0.0203	-0.0110	-0.0112	-0.0133	-0.0089
HC SE	(0.0053)***	(0.0058)***	(0.0055)**	(0.0055)**	(0.0047)***	(0.0049)*
Conley SE (70km)	(0.0066)***	(0.0067)***	(0.0056)**	(0.0055)**	(0.0057)**	(0.0048)*
Controls						
Core	No	Yes	Yes	Yes	No	Yes
Geography	No	No	Yes	Yes	No	Yes
Neighbors	No	No	No	Yes	No	Yes
τ_j Spatial Lag	No	No	No	No	Yes	Yes
Fit Statistics						
Num.Obs.	232	230	230	230	230	230
R2	0.074	0.141	0.370	0.389	0.309[a]	0.441[a]
Mean Dep. Var.	0.391	0.391	0.391	0.391	0.394	0.394

Table 4: Pooled Benchmark: Domain-Level Geographic Exposure and Tax Rates

Note: OLS estimates (ML in Models 5–6) of domain-average distance ratio on nominal tax rate. Full results in Table A.43. SEs in parentheses: (i) HC1 (ML SEs in 5–6); (ii) Conley (70 km) on capital coordinates, computed via two-step filtered-residual approximation (Anselin and Rey 2014, pp. 34–36); stars reflect that estimator's p-value. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1. [a] Nagelkerke pseudo-R2

To identify the exit channel, I interact geographic exposure with $\hat{y}_{-j}$, a discrete moderator for the quality of neighboring domains' outside option:

$$\tau_j = \alpha_5 + \beta_5\left[\ln\left(\hat{\theta}_j\right) \times \hat{y}_{-j}\right] + \bar{\beta}_5 \ln\left(\hat{\theta}_j\right) + \tilde{\beta}_5 \hat{y}_{-j} + X_j' \boldsymbol{\gamma}_5 + \varepsilon_j$$

I approximate the outside-option quality y_{-j} using neighboring domains' demand for labor. Rulers confronting labor shortages adopted measures to attract workers, such as land redistribution, provision of tools and rations, and investments in irrigation and other public goods (Hyugano 1957; Miyazaki 1995). Incentives to court migrants intensified with the severity of labor scarcity. In the Kantō (eastern) region, where shortages persisted throughout the period, domains implemented peasant in-migration (recruitment) policies (iri-byakushō seisaku) that offered tax exemptions, food and seed grants, and other settlement inducements to draw cultivators from neighboring territories (Gorai 1950). I construct $\hat{y}_{-j}$ as a binary indicator that equals "Low" when the average labor-to-productivity ratio of neighboring domains is *above* its median, and "High" otherwise.[25] The average is computed by weighting each neighbor by the number of domain j's villages for which that neighbor's capital is the nearest neighboring capital, and each neighbor's labor-to-productivity ratio is its population divided by its potential wet-rice yield from Galor and Özak (2016).[26] $\hat{y}_{-j} = \text{High}$ therefore marks domains whose villages face neighbors with greater labor-absorption capacity and thus stronger economic incentives to extend favorable treatment to potential migrants. In the interaction model, average rice yield and territory size of neighbors are absorbed into $\hat{y}_{-j}$ and dropped from X_j.

Figure 8 reports the main result. Across all specifications, the negative association between geographic exposure and tax rates is concentrated in domains with high $\hat{y}_{-j}$ (neighbors with low labor-to-potential-yield ratio, and thus strong demand for in-migrants). Once geographic characteristics are controlled (Models 3 and 4), the negative association attains statistical significance only for this high-$\hat{y}_{-j}$ group; for domains facing labor-abundant neighbors, the coefficients are small and statistically insignificant. Consistent with Proposition 3, geographic exposure constrains taxation only where the outside option is attractive enough to make exit credible. This asymmetry is the signature that distinguishes exit from state-capacity decay: remoteness-driven extractive slack would not vary with neighboring rulers' labor demand, whereas the exit threat, and its fiscal concession, should bind only where neighbors can credibly absorb migrants. The pattern replicates with population density as an alternative proxy for outside-option quality (Appendix E.3.3). Reconstructing $\hat{y}_{-j}$ with a fixed neighbor set (all domains within 100 km of the focal domain, independent of the nearest-capital weighting) yields the same high/low asymmetry and significance pattern, and a Hainmueller-Mummolo-Xu kernel-smoothed marginal-effect plot on the continuous moderator shows the negative association holds across the full support of neighbor labor-absorption capacity (Appendix E.3.4).

[25]Hainmueller, Mummolo, and Xu (2019) show that continuous multiplicative-interaction models rest on a linear-interaction assumption that frequently fails in practice, and recommend replacing the continuous moderator with a discrete one precisely for settings like this, in which the moderating relationship is inherently nonlinear.

[26]I use the potential yield measure under irrigated conditions with medium inputs.

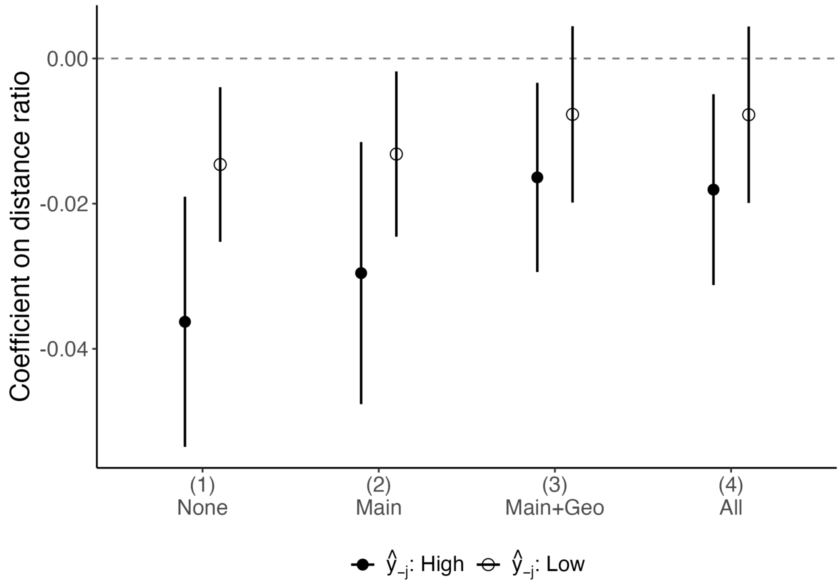


Figure 8: The Tax-Exposure Link Strengthens with Outside-Option Quality

Note: OLS coefficients on $\ln\left(\hat{\theta}_j\right) \times \hat{y}_{-j}$, where $\hat{y}_{-j}$ is above/below the median neighbor labor-to-potential-yield ratio. Filled: High; open: Low. 95% HC CIs. Controls cumulative: (1) none; (2) domain; (3) +geography; (4) +neighbor. Full table in Table A.44.

3.5.3. Additional Analyses

Two further analyses clarify what the geographic-exposure measure captures. First, simple remoteness from the home capital, the alternative predictor implied by state-capacity decay (Sng and Moriguchi 2014), does not explain the pattern: adding the average log distance to the home capital as a control leaves the geographic-exposure coefficient negative (Appendix E.2.4), and replacing the distance ratio with the remoteness measure renders any negative association insignificant once geographic controls enter (Appendix E.3.1). Second, distance to the nearest border, the alternative predictor if exit were about reaching foreign land rather than rival authority, likewise does not predict tax rates (Appendix E.3.2),[27] mirroring the village-level pattern from Section 3.4. Access to alternative authority, not mere remoteness or border proximity, is what constrains extraction.

[27]The boundary data were constructed following the procedure described in (Honda, Natsume, and Nemoto 2022).

4. Conclusion

This paper shows that access to alternative authority — the combination of geographic proximity to rival rulers and the quality of the outside option those rivals offer — determines whether subjects exit or fight. The empirical analysis proceeds in three parts. First, tenryō conversions varied the outside option, causing nearby exit to spike while fighting did not, and a grid-year panel confirms that greater geographic exposure predicts higher exit frequency. Second, among villages that did revolt, greater geographic exposure shifted the chosen form of resistance toward exit over confrontation. Third, the exit mechanism left a fiscal trace: geographic exposure predicted lower domain tax rates, and this association held only where neighboring domains offered attractive outside options, consistent with exit being credible only when both conditions are met.

Exit was not passive flight; it was a bargaining strategy that relied on shelter, mediation, and recognition from outside authorities. What enables exit as a distinct form of resistance is alternative authority, not mere border proximity or local disorder. Where rival rulers are willing and able to absorb migrants, the menu of resistance expands beyond direct confrontation, and rulers who ignore it risk losing their tax base.

The logic extends beyond Tokugawa Japan. Medieval European lordships competing for peasant tenants (Bailey 2014), precolonial African polities losing subjects to neighboring kingdoms, and modern federal systems share the same structure: fragmented authority with cross-jurisdictional allegiance shifts. In each, geographic access to a rival jurisdiction and its willingness to absorb newcomers should shape whether exit becomes viable. Whether the premise scales into the institutional outcomes those literatures emphasize lies beyond these data. The paper supplies direct community-level evidence for a premise load-bearing across the Great Divergence literature, fiscal-federalism theory, and the cartel theory of the territorial state: political fragmentation gives subjects usable leverage over rulers through competing authorities. The Tokugawa peasantry is a hard case for this premise (hegemonically constrained, administratively bound, and legally prohibited from migration), making the evidence a conservative test.

Data Availability Statement

Replication data and code will be deposited in the International Organization Dataverse (https://dataverse.harvard.edu/dataverse/IOJ) upon conditional acceptance. The full pipeline reproduces every table and figure from the raw inputs documented in the codebook.

Ethical Statement

This research uses publicly available historical records and archival data on Tokugawa-era Japan and does not involve human subjects. No IRB review was required.

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